

SNITCH FASHION SALES ANALYSIS REPORT

1. Project Overview

The Snitch Fashion Sales Analysis project focuses on analyzing sales data using Python and Power BI. Python was used for data cleaning and preprocessing, while Power BI was used for data modelling and dashboard creation. The project provides insights into sales trends, profitability, customer segments and product performance through interactive visualizations, helping businesses make informed decisions and improve overall performance.

Project Objectives

- To analyze the overall sales and profit performance of Snitch Fashion products.
- To identify top-performing product categories and customer segments based on sales data.
- To compare sales and profitability across different cities and regions.
- To analyze monthly and yearly sales trends and evaluate the impact of discounts.
- To develop an interactive Power BI dashboard for effective business decision-making.

2. Data Sources

Source Description and Timeline:

<https://drive.google.com/file/d/1ktB6TRrWhrPEvYEF0QBloPtAEvC2mZad/view?usp=sharing>

Domain:

Fashion Retail / E-Commerce Sales Analytics.

3.Problem Statement

Snitch Fashion generates a large amount of sales data, making manual analysis difficult and time-consuming. Therefore, this project aims to analyze the data and create an interactive Power BI dashboard to provide meaningful insights and support better business decisions.

4.Attribute (Column/Feature) Details:

S.No	Attribute Name	Description	Data Type
1	Order ID	Unique identification number for each order.	Whole Number
2	Customer Name	Name of customer who purchased products.	Text
3	Product Category	Category of products.	Text
4	Product Name	Name of product.	Text
5	Units Sold	Number of units sold.	Decimal Number
6	Unit Price	Price of one unit.	Decimal Number
7	Discount%	Discount percentage offered.	Decimal Number
8	Sales Amount	Total amount generated from sales.	Decimal Number
9	Order Date	Date of purchase.	Date

10	City	City where order was placed.	Text
11	Segment	Customer segment(B2B/B2C).	Text
12	Profit	Profit earned from each order.	Decimal Number
13	Month	Month extracted from Order Date.	Whole Number
14	Year	Year extracted from Order Date.	Whole Number
15	Total Sales	Overall sales amount.	Decimal Number
16	Region	Region corresponding to city.	Text

5.Tools & Technologies

- Python - Used for data cleaning and preprocessing.
- Pandas - Used for data manipulation and handling missing values.
- NumPy - Used for numerical computations.
- Matplotlib and Seaborn - Used for exploratory data analysis and visualization.
- DAX - Used for calculated columns and measures.
- Power BI - Used for dashboard creation and data visualization.

6.Data Pre – Processing (Python)

Task Performed:

Python was used for data cleaning and preprocessing, including handling missing values, removing duplicate records, converting data types and preparing the dataset for further analysis.

7.Data Modelling and DAX (Power BI)

Data Modelling:

Power BI was used to create relationships between tables for effective analysis.

Calculated Columns and DAX Measures:

DAX functions were used to create calculated columns and measures for analyzing sales and profit performance.

SCREENSHOT OF DATA MODELLING AND DAX

Calculated Columns and DAX Measures: Total Sales

Table: Snitch Fashion Sales (1,101 rows) Column: Total Sales (0 distinct values)

Update available (click to download)

Total Profit

Snitch Fashion Sales Analysis • Last saved: Today at 2:56 PM

File Home Help Table tools Measure tools

Name Snitch Fashion Sales

Structure Relationships Calculations Calendars

1 Total Profit = SUM('Snitch Fashion Sales'[Profit])

Order_ID	Customer_Name	Product_Category	Product_Name	Units_Sold	Unit_Price	Discount_%	Sales_Amount	Order_Date	City
1005	Samuel Miller	T-Shirts	Crop Top	1.94	2691.72	64.65	1846.64	45265	Mumbai
1026	Colton Baxter	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45647	Ahmedab
1040	Angel Nelson	Jackets	Denim Jacket	1.94	2691.72	64.65	1846.64	45611	Ahmedab
1041	Teresa Chavez	Jackets	Puffer Coat	1.94	2691.72	64.65	1846.64	45723	Ahmedab
1035	Laura Porter	Accessories	Belts	1.94	2691.72	64.65	1846.64	45661	Mumbai
1099	Valerie Walker	Shoes	Loafers	1.94	2691.72	64.65	1846.64	45344	Pune
1119	Elizabeth Vega	Jeans	Ripped Denim	1.94	2691.72	64.65	1846.64	45791	Mumbai
1181	John Barber	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45732	Mumbai
1032	Stephanie Booth	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45847	Ahmedab
1224	Shelby Lewis	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45737	Ahmedab
1290	Dylan Newton	T-Shirts	Graphic Tee	1.94	2691.72	64.65	1846.64	45657	Ahmedab
1309	Tina Brown	Accessories	Caps	1.94	2691.72	64.65	1846.64	45719	Pune
1310	Thomas Smith	Dresses	Maxi Dress	1.94	2691.72	64.65	1846.64	45720	Pune
1312	Wendy Garcia	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45806	Pune
1324	Mike Rhodes	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45638	Mumbai
1028	John Taylor	Accessories	Watches	1.94	2691.72	64.65	1846.64	45240	Ahmedab
1361	Hannah Ball	T-Shirts	Classic Tee	1.94	2691.72	64.65	1846.64	45660	Mumbai
1395	Andrew Wood	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45393	Mumbai
1408	William Golden	Shoes	Boots	1.94	2691.72	64.65	1846.64	45539	Mumbai
1427	Angela Hall	Shoes	Boots	1.94	2691.72	64.65	1846.64	45826	Mumbai

Table: Snitch Fashion Sales (1,101 rows) Column: Total Profit (0 distinct values)

Update available (click to download)

Data

Search

Snitch Fashion Sales

City

Customer_Name

Discount_%

Month

Order_Date

Order_ID

Product_Category

Product_Name

Profit

Region

Sales_Amount

Segment

Total Profit

Total Sales

Total_Sales

Unit_Price

Total Unit Sold

Snitch Fashion Sales Analysis • Last saved: Today at 2:56 PM

File Home Help Table tools Measure tools

Name Snitch Fashion Sales

Structure Relationships Calculations Calendars

1 Total Unit Sold = SUM('Snitch Fashion Sales'[Units_Sold])

Order_ID	Customer_Name	Product_Category	Product_Name	Units_Sold	Unit_Price	Discount_%	Sales_Amount	Order_Date	City
1005	Samuel Miller	T-Shirts	Crop Top	1.94	2691.72	64.65	1846.64	45265	Mumbai
1026	Colton Baxter	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45647	Ahmedab
1040	Angel Nelson	Jackets	Denim Jacket	1.94	2691.72	64.65	1846.64	45611	Ahmedab
1041	Teresa Chavez	Jackets	Puffer Coat	1.94	2691.72	64.65	1846.64	45723	Ahmedab
1035	Laura Porter	Accessories	Belts	1.94	2691.72	64.65	1846.64	45661	Mumbai
1099	Valerie Walker	Shoes	Loafers	1.94	2691.72	64.65	1846.64	45344	Pune
1119	Elizabeth Vega	Jeans	Ripped Denim	1.94	2691.72	64.65	1846.64	45791	Mumbai
1181	John Barber	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45732	Mumbai
1032	Stephanie Booth	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45847	Ahmedab
1224	Shelby Lewis	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45737	Ahmedab
1290	Dylan Newton	T-Shirts	Graphic Tee	1.94	2691.72	64.65	1846.64	45657	Ahmedab
1309	Tina Brown	Accessories	Caps	1.94	2691.72	64.65	1846.64	45719	Pune
1310	Thomas Smith	Dresses	Maxi Dress	1.94	2691.72	64.65	1846.64	45720	Pune
1312	Wendy Garcia	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45806	Pune
1324	Mike Rhodes	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45638	Mumbai
1028	John Taylor	Accessories	Watches	1.94	2691.72	64.65	1846.64	45240	Ahmedab
1361	Hannah Ball	T-Shirts	Classic Tee	1.94	2691.72	64.65	1846.64	45660	Mumbai
1395	Andrew Wood	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45393	Mumbai
1408	William Golden	Shoes	Boots	1.94	2691.72	64.65	1846.64	45539	Mumbai
1427	Angela Hall	Shoes	Boots	1.94	2691.72	64.65	1846.64	45826	Mumbai

Table: Snitch Fashion Sales (1,101 rows) Column: Total Unit Sold (0 distinct values)

Update available (click to download)

Data

Search

Snitch Fashion Sales

City

Customer_Name

Discount_%

Month

Order_Date

Order_ID

Product_Category

Product_Name

Profit

Region

Sales_Amount

Segment

Total Profit

Total Sales

Total Unit Sold

Total_Sales

Average Discount

Snitch Fashion Sales Analysis • Last saved: Today at 2:56 PM

File Home Help Table tools Measure tools

Name: Snitch Fashion Sales

Structure: Manage relationships, New measure, Quick measure, New column, New table, Mark as date table, Calculations, Calendars

1 Average Discount - $AVERAGE('Snitch Fashion Sales'[Discount_ \%])$

Order_ID	Customer_Name	Product_Category	Product_Name	Units_Sold	Unit_Price	Discount_ %	Sales_Amount	Order_Date	City
1005	Samuel Miller	T-Shirts	Crop Top	1.94	2691.72	64.65	1846.64	45265	Mumbai
1026	Colton Baxter	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45647	Ahmedab
1040	Angel Nelson	Jackets	Denim Jacket	1.94	2691.72	64.65	1846.64	45611	Ahmedab
1041	Teresa Chavez	Jackets	Puffer Coat	1.94	2691.72	64.65	1846.64	45723	Ahmedab
1035	Laura Porter	Accessories	Belts	1.94	2691.72	64.65	1846.64	45661	Mumbai
1099	Valerie Walker	Shoes	Loafers	1.94	2691.72	64.65	1846.64	45344	Pune
1119	Elizabeth Vega	Jeans	Ripped Denim	1.94	2691.72	64.65	1846.64	45791	Mumbai
1181	John Barber	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45732	Mumbai
1032	Stephanie Booth	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45847	Ahmedab
1224	Shelby Lewis	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45737	Ahmedab
1298	Dylan Newton	T-Shirts	Graphic Tee	1.94	2691.72	64.65	1846.64	45657	Ahmedab
1309	Tina Brown	Accessories	Caps	1.94	2691.72	64.65	1846.64	45719	Pune
1310	Thomas Smith	Dresses	Maxi Dress	1.94	2691.72	64.65	1846.64	45720	Pune
1312	Wendy Garcia	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	45806	Pune
1324	Mike Rhodes	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	45638	Mumbai
1028	John Taylor	Accessories	Watches	1.94	2691.72	64.65	1846.64	45240	Ahmedab
1361	Hannah Ball	T-Shirts	Classic Tee	1.94	2691.72	64.65	1846.64	45660	Mumbai
1395	Andrew Wood	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	45393	Mumbai
1408	William Golden	Shoes	Boots	1.94	2691.72	64.65	1846.64	45539	Mumbai
1427	Anqela Hall	Shoes	Boots	1.94	2691.72	64.65	1846.64	45826	Mumbai

Table: Snitch Fashion Sales (1,101 rows) Column: Average Discount (0 distinct values)

Data: Snitch Fashion Sales, Average Discount, City, Customer_Name, Discount_ %, Month, Order_Date, Order_ID, Product_Category, Product_Name, Profit, Region, Sales_Amount, Segment, Total Profit, Total Sales, Total Unit Sold

Sales Target

Snitch Fashion Sales Analysis • Last saved: Yesterday at 11:26 PM

File Home Help Table tools Measure tools

Name: Sales Target

Format: General, Data category: Uncategorized

Home table: Snitch Fashion Sales

Structure: Formatting, Properties, Calculations

1 Sales Target - $[Total Sales] * 1.2$

Order_ID	Customer_Name	Product_Category	Product_Name	Units_Sold	Unit_Price	Discount_ %	Sales_Amount	Order_Date	City
1005	Samuel Miller	T-Shirts	Crop Top	1.94	2691.72	64.65	1846.64	05-12-2023	Mumbai
1026	Colton Baxter	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	21-12-2024	Ahmedab
1040	Angel Nelson	Jackets	Denim Jacket	1.94	2691.72	64.65	1846.64	15-11-2024	Ahmedab
1041	Teresa Chavez	Jackets	Puffer Coat	1.94	2691.72	64.65	1846.64	07-03-2025	Ahmedab
1035	Laura Porter	Accessories	Belts	1.94	2691.72	64.65	1846.64	04-01-2025	Mumbai
1099	Valerie Walker	Shoes	Loafers	1.94	2691.72	64.65	1846.64	22-02-2024	Pune
1119	Elizabeth Vega	Jeans	Ripped Denim	1.94	2691.72	64.65	1846.64	14-05-2025	Mumbai
1181	John Barber	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	16-03-2025	Mumbai
1032	Stephanie Booth	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	09-07-2025	Ahmedab
1224	Shelby Lewis	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	21-03-2025	Ahmedab
1298	Dylan Newton	T-Shirts	Graphic Tee	1.94	2691.72	64.65	1846.64	31-12-2024	Ahmedab
1309	Tina Brown	Accessories	Caps	1.94	2691.72	64.65	1846.64	03-03-2025	Pune
1310	Thomas Smith	Dresses	Maxi Dress	1.94	2691.72	64.65	1846.64	04-03-2025	Pune
1312	Wendy Garcia	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	29-05-2025	Pune
1324	Mike Rhodes	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	12-12-2024	Mumbai
1028	John Taylor	Accessories	Watches	1.94	2691.72	64.65	1846.64	10-11-2023	Ahmedab
1361	Hannah Ball	T-Shirts	Classic Tee	1.94	2691.72	64.65	1846.64	03-01-2025	Mumbai
1395	Andrew Wood	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	11-04-2024	Mumbai
1408	William Golden	Shoes	Boots	1.94	2691.72	64.65	1846.64	04-09-2024	Mumbai
1427	Anqela Hall	Shoes	Boots	1.94	2691.72	64.65	1846.64	18-06-2025	Mumbai

Table: Snitch Fashion Sales (1,101 rows) Column: Sales Target (0 distinct values)

Data: Snitch Fashion Sales, Average Discount, City, Customer_Name, Discount_ %, Month, Order_Date, Order_ID, Product_Category, Product_Name, Profit, Region, Sales_Amount, Segment, Sales Target, Total Profit, Total Sales

Cleaned Dataset Table

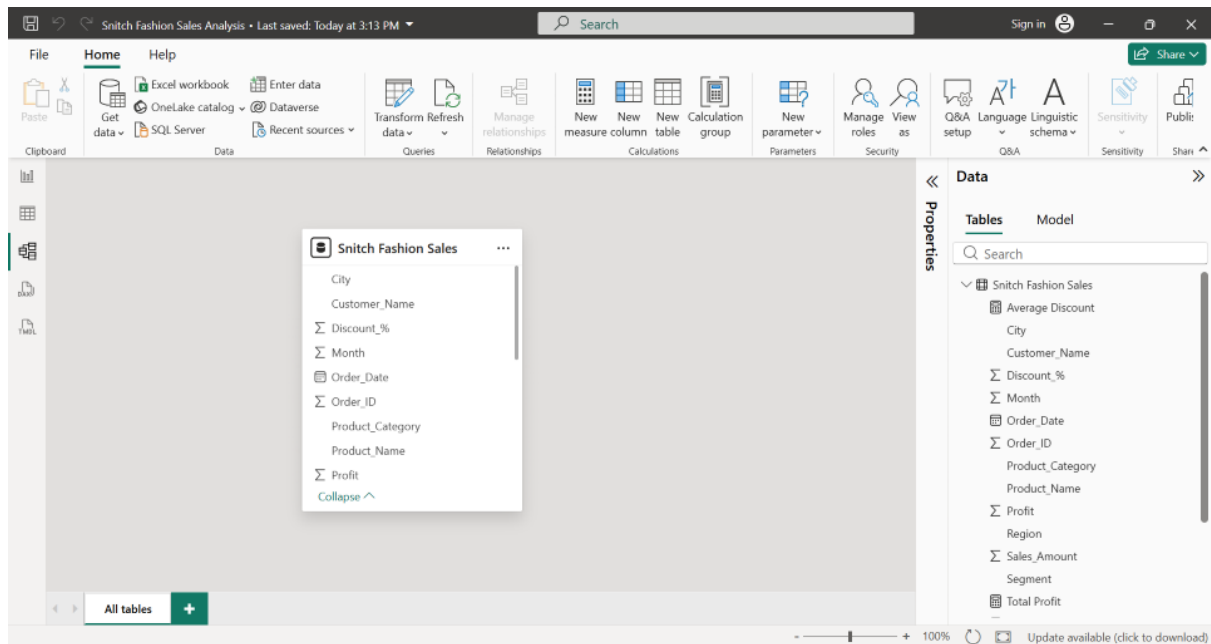
Snitch Fashion Sales Analysis • Last saved: Today at 4:03 PM													
Search													
File Home Help Table tools Column tools													
Name Order_ID Format Whole number Summarization Count Data category Uncategorized Sort by column Data groups Manage relationships New column													
Structure Formatting Properties Sort Groups Relationships Calculations													
Order_ID Customer_Name Product_Category Product_Name Units_Sold Unit_Price Discount_% Sales_Amount Order_Date City Segment Profit Month													
1005	Samuel Miller	T-Shirts	Crop Top	1.94	2691.72	64.65	1846.64	05-12-2023	Mumbai	B2B	1477.73		
1026	Colton Baxter	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	21-12-2024	Ahmedabad	B2B	194.77		
1040	Angel Nelson	Jackets	Denim Jacket	1.94	2691.72	64.65	1846.64	15-11-2024	Ahmedabad	B2C	693.6		
1041	Teresa Chavez	Jackets	Puffer Coat	1.94	2691.72	64.65	1846.64	07-03-2025	Ahmedabad	B2C	983.06		
1035	Laura Porter	Accessories	Belts	1.94	2691.72	64.65	1846.64	04-01-2025	Mumbai	B2B	1539.59		
1099	Valerie Walker	Shoes	Loafers	1.94	2691.72	64.65	1846.64	22-02-2024	Pune	B2C	300.02		
1119	Elizabeth Vega	Jeans	Ripped Denim	1.94	2691.72	64.65	1846.64	14-05-2025	Mumbai	B2C	142.69		
1181	John Barber	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	16-03-2025	Mumbai	B2B	2564.06		
1032	Stephanie Booth	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	09-07-2025	Ahmedabad	B2C	1941.65		
1224	Shelby Lewis	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	21-03-2025	Ahmedabad	B2C	2219.56		
1290	Dylan Newton	T-Shirts	Graphic Tee	1.94	2691.72	64.65	1846.64	31-12-2024	Ahmedabad	B2C	2490.23		
1309	Tina Brown	Accessories	Caps	1.94	2691.72	64.65	1846.64	03-03-2025	Pune	B2B	481.45		
1310	Thomas Smith	Dresses	Maxi Dress	1.94	2691.72	64.65	1846.64	04-03-2025	Pune	B2B	1516.35		
1312	Wendy Garcia	Accessories	Sunglasses	1.94	2691.72	64.65	1846.64	29-05-2025	Pune	B2B	646.16		
1324	Mike Rhodes	Jackets	Bomber Jacket	1.94	2691.72	64.65	1846.64	12-12-2024	Mumbai	B2C	610.52		
1028	John Taylor	Accessories	Watches	1.94	2691.72	64.65	1846.64	10-11-2023	Ahmedabad	B2B	2502.73		
1361	Hannah Ball	T-Shirts	Classic Tee	1.94	2691.72	64.65	1846.64	03-01-2025	Mumbai	B2C	528.8		
1395	Andrew Wood	Dresses	Wrap Dress	1.94	2691.72	64.65	1846.64	11-04-2024	Mumbai	B2B	357.08		
1408	William Golden	Shoes	Boots	1.94	2691.72	64.65	1846.64	04-09-2024	Mumbai	B2B	2305.32		
1427	Anqela Hall	Shoes	Boots	1.94	2691.72	64.65	1846.64	18-06-2025	Mumbai	B2C	2795.17		

Table: Snitch Fashion Sales (1,101 rows) Column: Order_ID (1,064 distinct values)

Snitch Fashion Sales Analysis • Last saved: Today at 4:03 PM													
Search													
File Home Help Table tools Column tools													
Name Order_ID Format Whole number Summarization Count Data category Uncategorized Sort by column Data groups Manage relationships New column													
Structure Formatting Properties Sort Groups Relationships Calculations													
Product_Category Product_Name Units_Sold Unit_Price Discount_% Sales_Amount Order_Date City Segment Profit Month Year Total_Sales Region													
ts	Crop Top	1.94	2691.72	64.65	1846.64	05-12-2023	Mumbai	B2B	1477.73	12	2023	5223.37	West
sories	Sunglasses	1.94	2691.72	64.65	1846.64	21-12-2024	Ahmedabad	B2B	194.77	12	2024	5223.37	West
ts	Denim Jacket	1.94	2691.72	64.65	1846.64	15-11-2024	Ahmedabad	B2C	693.6	11	2024	5223.37	West
ts	Puffer Coat	1.94	2691.72	64.65	1846.64	07-03-2025	Ahmedabad	B2C	983.06	3	2025	5223.37	West
sories	Belts	1.94	2691.72	64.65	1846.64	04-01-2025	Mumbai	B2C	1539.59	1	2025	5223.37	West
ts	Loafers	1.94	2691.72	64.65	1846.64	22-02-2024	Pune	B2B	300.02	2	2024	5223.37	West
ts	Ripped Denim	1.94	2691.72	64.65	1846.64	14-05-2025	Mumbai	B2C	142.69	5	2025	5223.37	West
ts	Bomber Jacket	1.94	2691.72	64.65	1846.64	16-03-2025	Mumbai	B2B	2564.06	3	2025	5223.37	West
sories	Sunglasses	1.94	2691.72	64.65	1846.64	09-07-2025	Ahmedabad	B2C	1941.65	7	2025	5223.37	West
es	Wrap Dress	1.94	2691.72	64.65	1846.64	21-03-2025	Ahmedabad	B2C	2219.56	3	2025	5223.37	West
ts	Graphic Tee	1.94	2691.72	64.65	1846.64	31-12-2024	Ahmedabad	B2C	2490.23	12	2024	5223.37	West
sories	Caps	1.94	2691.72	64.65	1846.64	03-03-2025	Pune	B2B	481.45	3	2025	5223.37	West
es	Maxi Dress	1.94	2691.72	64.65	1846.64	04-03-2025	Pune	B2B	1516.35	3	2025	5223.37	West
sories	Sunglasses	1.94	2691.72	64.65	1846.64	29-05-2025	Pune	B2B	646.16	5	2025	5223.37	West
ts	Bomber Jacket	1.94	2691.72	64.65	1846.64	12-12-2024	Mumbai	B2C	610.52	12	2024	5223.37	West
sories	Watches	1.94	2691.72	64.65	1846.64	10-11-2023	Ahmedabad	B2B	2502.73	11	2023	5223.37	West
ts	Classic Tee	1.94	2691.72	64.65	1846.64	03-01-2025	Mumbai	B2C	528.8	1	2025	5223.37	West
es	Wrap Dress	1.94	2691.72	64.65	1846.64	11-04-2024	Mumbai	B2B	357.08	4	2024	5223.37	West
ts	Boots	1.94	2691.72	64.65	1846.64	04-09-2024	Mumbai	B2B	2305.32	9	2024	5223.37	West
ts	Boots	1.94	2691.72	64.65	1846.64	18-06-2025	Mumbai	B2C	2795.17	6	2025	5223.37	West

Table: Snitch Fashion Sales (1,101 rows) Column: Order_ID (1,064 distinct values)

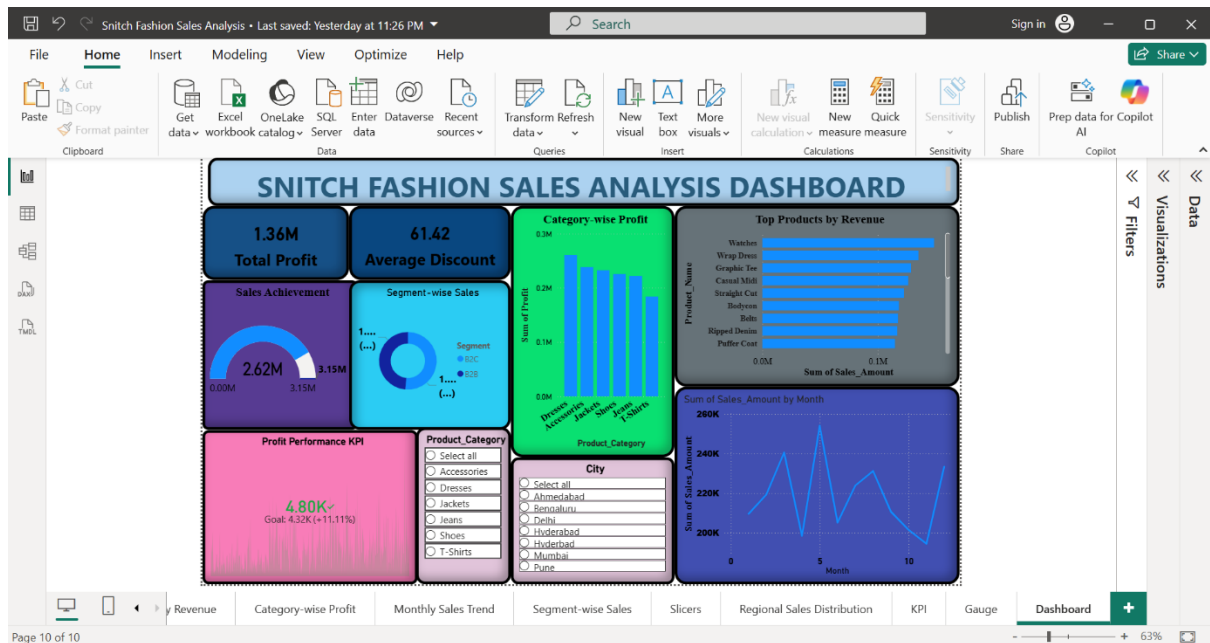
Data Modelling



8. Analysis and Visualization Dashboard (Features)

Power BI dashboards were created using KPI cards, Gauge charts, Donut charts, Column analyze sales performance, Profit trends, Products categories, customer segments and monthly sales patterns.

SCREENSHOT OF DASHBOARD:



9. Insights & Conclusion

Key Findings

- Sales performance varied across different months and years.
- Product categories contributed differently to overall profit and sales.
- B2B and B2C customer segments showed different purchasing patterns.
- Certain products generated higher sales compared to others.
- Discounts had a significant impact on profitability.
- Monthly sales trends revealed fluctuations in customer demand.

Analysis Insights

Descriptive Analysis

- Summarized the overall sales and profit performance of Snitch Fashion.
- Analyzed product categories, customer segments and monthly sales trends.
- Visualized sales patterns using various Power BI charts and KPIs.

Diagnostic Analysis

- Identified the factors affecting sales and profitability.
- Compared the performance of different product categories and customer segments.
- Examined the impact of discounts and monthly trends on business performance.

10. Conclusion

The Snitch Fashion Sales Analysis project successfully transformed raw data into meaningful business insights using Python and Power BI. The interactive dashboard provided meaningful insights into Sales performance, Profit trends, Product categories, Customer segments, Monthly sales patterns, enabling better decision-making and Supporting business growth.